Different Types of Matrix

Types of Matrices:

Diagonal Matrix: A square matrix where all the elements are zero except those on the main diagonal. That means a matrix $B = [b_{ij}]_{m \times m}$ is said to be a diagonal matrix if $b_{ij} = 0$, when $i \neq j$.

Example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

Scalar Matrix: A diagonal matrix is said to be a scalar matrix if all the elements in its principal diagonal are equal to some non-zero constant. A diagonal matrix is said to be a scalar matrix if its diagonal elements are equal, that is, a square matrix B = $[b_{ij}]_{n \times n}$ is said to be a scalar matrix if,

- $b_{ij} = 0$, when $i \neq j$
- $b_{ij} = k$, when $i = j$, for some constant k.

Example:

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Identity/Unit Matrix: An identity matrix is a square matrix that has 1's along the main diagonal and 0's everywhere else and denoted by I. A square matrix A = $[a_{ij}]_{n \times n}$ is an identity matrix if

- $a_{ij} = 1$ if $i = j$
- $a_{ij} = 0$ if $i \neq j$

Notation: I_n for $n \times n$ unit matrix. When the order is clear from the context, we simply write it as I.

Example:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Diagonal matrix:

$$M = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & -5 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Scalar matrix:

$$M = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Identity matrix:

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Triangular Matrix: A square matrix whose elements above or below the main diagonal are all zero. A square matrix A = $[a_{ij}]_{n \times n}$ be a triangular matrix if

- $a_{ij} = 0$ if $i < j$

  or

- $a_{ij} = 0$ if $i > j$

Example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 8 & 9 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

Upper Triangular Matrix: A square matrix whose elements below the main diagonal are all zero. A square matrix A = $[a_{ij}]_{n \times n}$ be a upper triangular matrix if

- $a_{ij} = 0$ if $i > j$

Example:

$$\begin{bmatrix} 1 & 8 & 7 \\ 0 & 1 & 8 \\ 0 & 0 & 3 \end{bmatrix}$$

Lower Triangular Matrix: A square matrix whose elements above the main diagonal are all zero. A square matrix A = $[a_{ij}]_{n \times n}$ be a lower triangular matrix if

- $a_{ij} = 0$ if $i < j$

Example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

DIAGONAL AND TRACE:

Only a square matrix posses the diagonal & the trace.

Example:

$$M = \begin{bmatrix} 3 & 1 & 2 & 0 & -1 \\ 0 & -2 & 1 & -1 & 2 \\ 2 & 1 & 2.7 & 2 & 0 \\ 1 & -1 & 1 & 0 & -2 \\ 6 & 4 & 3 & 2 & -8.5 \end{bmatrix}$$

1. Diag (M) = {3, -2, 2.7, 0, -8.5}
2. Tr (M) = 3 + (-2) + 2.7 + 0 + (-8.5) = -4.8

PERMUTATION MATRIX

A permutation matrix (P) looks very similar to an identity Matrix. A permutation Matrix (P):

i. Is a square matrix
ii. Consists of only 1's and 0's
iii. Each row must consist of a single 1
iv. Each column must consist of a single 1

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix P_1 Matrix P_2 Matrix P_3

Zero/Null/Void Matrix	Nonzero Matrix	Sparse Matrix	Dense Matrix
Each entry is zero (0)	At least one nonzero element	Number of Zeros > Number of Non-zeros	Number of Zeros < Number of Non-zeros

N.B: A Sparse/Dense matrix must have zero and nonzero elements as entries.

Example:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Zero matrix   Nonzero matrix

$$\begin{bmatrix} 1 & -4 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 1 \\ 9 & 9 & 0 \\ 6 & 6 & 1 \end{bmatrix}$$

Sparse matrix   Dense matrix

Exercise

1. Give example of the following matrices:
   i. Upper triangular matrix with order 4×4, 6×6 and 4×6.
   ii. Lower triangular matrix with order 3×3, 5×5 and 6×5.
   iii. Unit matrix with order 4×4, 6×6 and 4×6.
   iv. Permutation matrix with order 3×3, 5×5 and 6×5.
   v. Scalar matrix with order 4×4, 6×6 and 4×6.
   vi. Diagonal matrix with order 3×3, 5×5 and 6×5.
   vii. Row matrix with 13 entries.
   viii. Column matrix with 11 entries.
2. Find True or False from the following statements:
   a) Every diagonal matrix is a scalar matrix.

b) Every diagonal matrix is a unit matrix.

c) Every unit matrix is a scalar matrix.

d) Every unit matrix is a diagonal matrix.

e) Every scalar matrix is a diagonal matrix.

f) Every scalar matrix is a unit matrix.

3. What is the trace of -5I_{132}?
4. What is the trace of a zero matrix?
5. What is the trace of a square zero matrix?
6. 17 tr(I_{23}) – tr(22I_{17}) =?

Matrix Operations:

There are three types of matrix operations,

i. Scalar Multiplication of Matrix

ii. Matrix Addition & Subtraction

iii. Matrix Multiplication

1. **Scalar Multiplication:** It is defined with, "multiply each element in the matrix by the given scalar".

   Example: Let, $A = \begin{bmatrix} 21 & -2 \\ -5 & -1 \end{bmatrix}$, then multiplying A by a scalar 2. We get, $2\text{A} = \begin{bmatrix} 42 & -4 \\ -10 & -2 \end{bmatrix}$.

2. **Matrix Addition & Subtraction:** Two matrices can only be added or subtracted if and only if they have the same dimensions. Simply add (or subtract) the corresponding elements.

   Example: Let, $\text{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $\text{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$.

   Then, we get $\text{C} = \text{A} + \text{B} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$;

   Where, $\begin{cases} c_{11} = a_{11} + b_{11}; & c_{12} = a_{12} + b_{12} \\ c_{21} = a_{21} + b_{21}; & c_{22} = a_{22} + b_{22} \end{cases}$

   Problem 01: If, $\text{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 6 & 7 \end{bmatrix}$ and $\text{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ then find A+B.

Solution: $A+B=\begin{bmatrix}1&2\\3&4\\6&7\end{bmatrix}+\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}=\begin{bmatrix}1+1&2+0\\3+1&4+1\\6+0&7+1\end{bmatrix}=\begin{bmatrix}2&2\\4&5\\6&8\end{bmatrix}$.

Problem 02: If, $A=\begin{bmatrix}-4&3\\1&9\end{bmatrix}$ and $B=\begin{bmatrix}5&1&0\\1&2&0\\0&4&1\end{bmatrix}$ then find A+B.

Solution: The addition is undefined.

3. **Matrix Multiplication:** Condition of matrix multiplication is, "Matrix multiplication between two matrices will be possible if and only if the number of columns of the first matrix is equal to the number of rows of the second matrix".

   Let A and B are two matrix of order $m \times n$ and $p \times q$, then AB possible if and only if $n = p$ and the dimension of AB is $m \times q$.

   Properties of Matrix multiplication:

   i. (AB)C =A(BC)

   ii. A(B+C)=AB+AC

   iii. (B+C)A=BA+CA

   Example: If, $A=\begin{bmatrix}1&-2&1\\2&1&3\end{bmatrix}$ and $B=\begin{bmatrix}2&1\\3&2\\1&1\end{bmatrix}$ then find AB.

   Solution:

$$AB=\begin{bmatrix}1&-2&1\\2&1&3\end{bmatrix}\times\begin{bmatrix}2&1\\3&2\\1&1\end{bmatrix}$$

$$=\begin{bmatrix}1\times2+(-2)\times3+1\times1 & 1\times1+(-2)\times2+1\times1\\2\times2+1\times3+3\times1 & 2\times1+1\times2+3\times1\end{bmatrix}$$

$$\boxed{AB=\begin{bmatrix}-3&-2\\10&7\end{bmatrix}}$$

Idempotent Matrix: The matrix A is idempotent matrix if and only if $A^2=A$. For this, product A^2 to be defined, A must necessarily be a square matrix.

Example:

$$\begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix}; \quad \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

Involuntary matrix: If $A^2 = I$ or $A^{-1} = A$, then the matrix A is said to be an involuntary matrix.

Example:

$$\begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

Nilpotent matrix: A nilpotent matrix is a square matrix N such that $N^k = 0$ for some positive integer k. The smallest such k is called the **index** of N, sometimes the **degree** of N.

Example:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Exercise

Short question

- When can you subtract 1 matrix from another ?
- If A is of the order of 6×9 & B is of the order 9×6 , then what's the order of $A+B$=?
- L is 4×29, M is 27×4 order matrix. What will be the dimension of ML? Dimension of LM ?
- B is 14×19 & A is 19×14. Does $BA + 14I_{19}$ exist?
- P & Q posses the same dimension. When does PQ not exist ?
- P & Q posses the same dimension. When does PQ must exist ?
- Can a zero matrix be added to any matrix ?
- Can a unit matrix be added to any matrix ?
- Dimensions of L, M & N are 7×9, 9×13, 13×17 & respectively. Dimension of LMN ? Of MNL ?

Broad question

1. $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ -1 & 0 & 3 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 3 & 0 \\ 1 & 7 & -5 \\ 3 & 2 & 0 \end{pmatrix}$

Compute

$(a).\, A+B, A-B \quad (b).\, AB, BA \quad (c).\, A^2B, B^2A$

2. $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & -1 \\ 2 & -1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} -1 & 2 & 1 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

Then find $\boxed{4A + 5AB - 3I_3}$